```r
# Numeric vector
n <- c(10, 20, 30)
print(n)
print(typeof(n))

# Character vector
ch <- c("Apple", "Banana", "Mango")
print(ch)
print(typeof(ch))

# Logical vector
l <- c(TRUE, FALSE, TRUE)
print(l)
print(typeof(l))
```

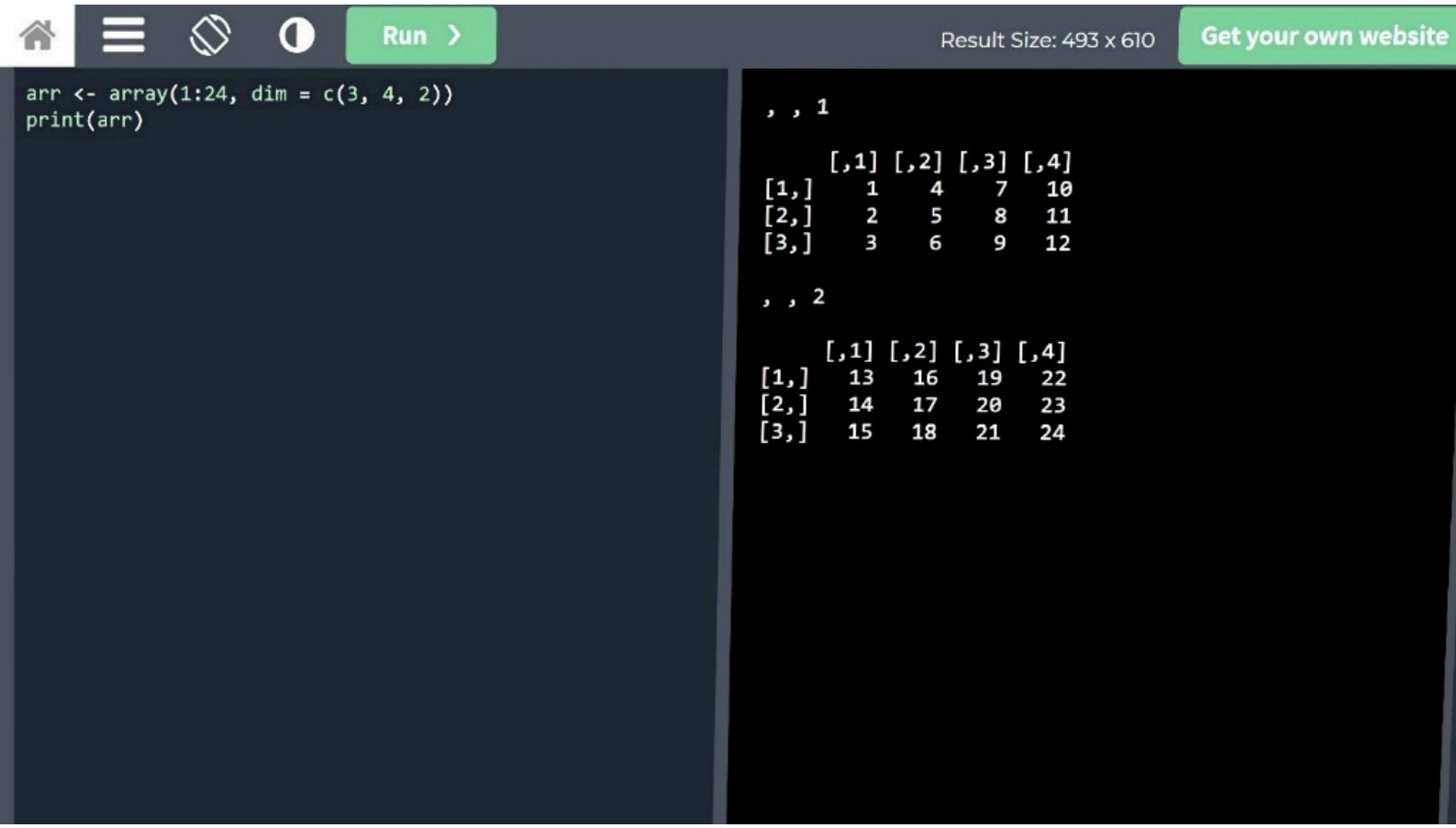
```
[1] 10 20 30
[1] "double"
[1] "Apple"  "Banana" "Mango"
[1] "character"
[1]  TRUE FALSE  TRUE
[1] "logical"
```

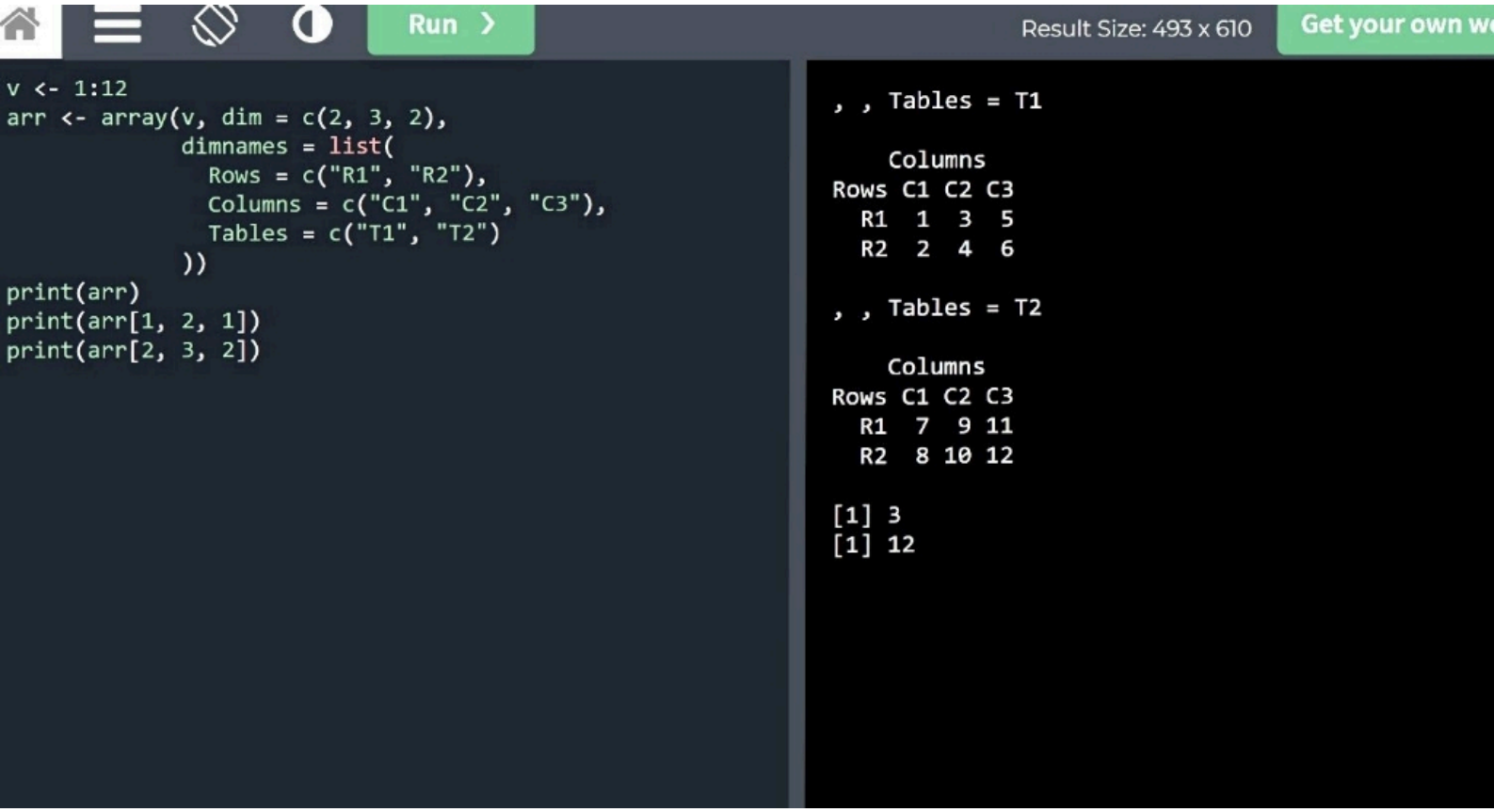
```r
# 5x4 matrix
A <- matrix(1:20, nrow=5, ncol=4, byrow=TRUE,
            dimnames=list(paste0("R",1:5),
paste0("C",1:4)))
print(A)

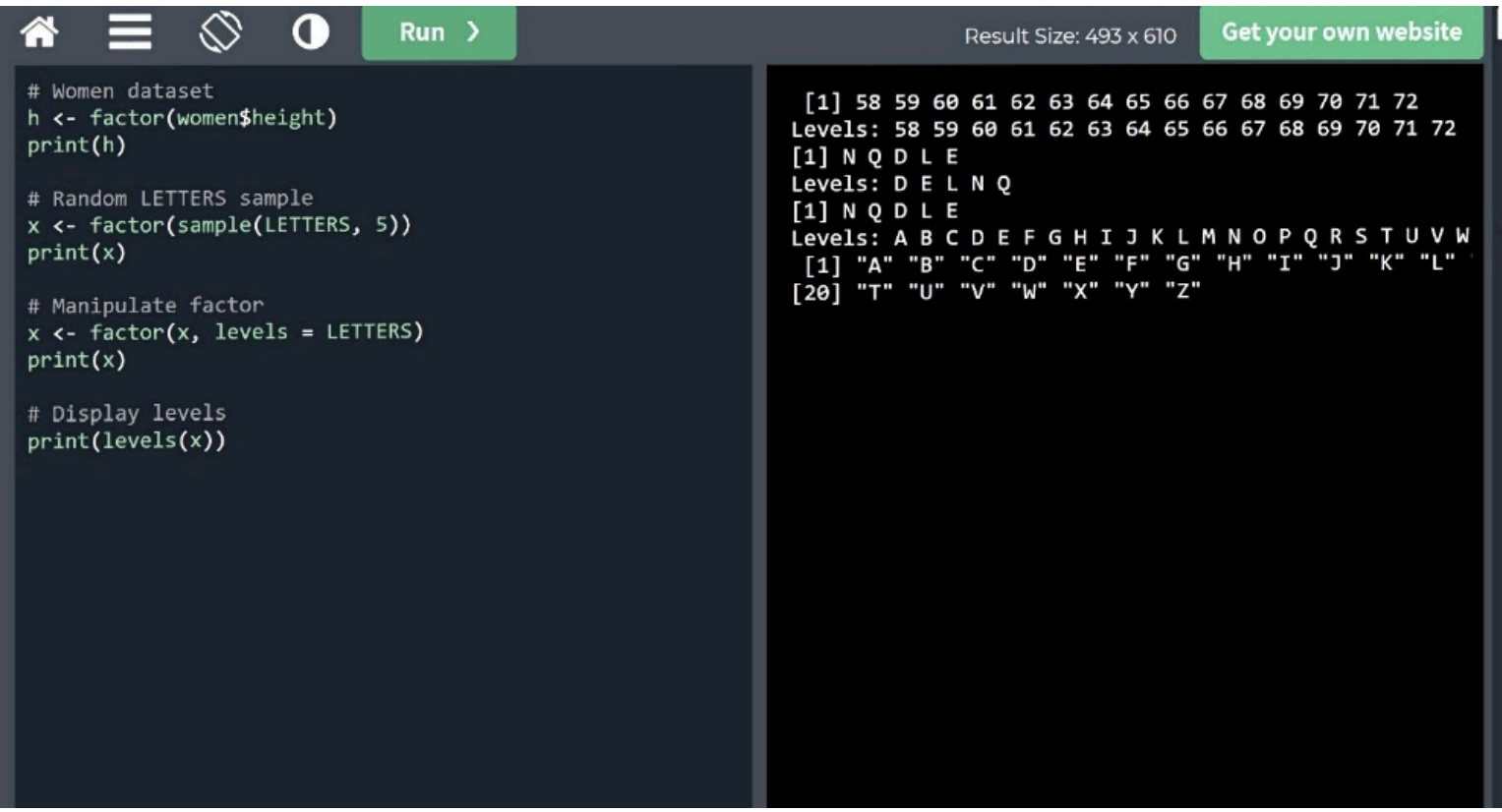
# 3x3 matrix
B <- matrix(1:9, nrow=3, ncol=3, byrow=FALSE,
            dimnames=list(paste0("R",1:3),
paste0("C",1:3)))
print(B)

# 2x2 matrix
C <- matrix(1:4, nrow=2, ncol=2, byrow=TRUE,
            dimnames=list(paste0("R",1:2),
paste0("C",1:2)))
print(C)
```

```
   C1 C2 C3 C4
R1  1  2  3  4
R2  5  6  7  8
R3  9 10 11 12
R4 13 14 15 16
R5 17 18 19 20
   C1 C2 C3
R1  1  4  7
R2  2  5  8
R3  3  6  9
   C1 C2
R1  1  2
R2  3  4
```


Run
Result Size: 493 x 610
Get your own website
arr <- array(1:24, dim = c(3, 4, 2))
print(arr)
, , 1
[,1] [,2] [,3] [,4]
[1,] 1 4 7 10
[2,] 2 5 8 11
[3,] 3 6 9 12
, , 2
[,1] [,2] [,3] [,4]
[1,] 13 16 19 22
[2,] 14 17 20 23
[3,] 15 18 21 24


Run
Result Size: 493 x 610
Get your own w
v <- 1:12
arr <- array(v, dim = c(2, 3, 2),
dimnames = list(
Rows = c("R1", "R2"),
Columns = c("C1", "C2", "C3"),
Tables = c("T1", "T2")
))
print(arr)
print(arr[1, 2, 1])
print(arr[2, 3, 2])
, , Tables = T1
Columns
Rows C1 C2 C3
R1 1 3 5
R2 2 4 6
, , Tables = T2
Columns
Rows C1 C2 C3
R1 7 9 11
R2 8 10 12
[1] 3
[1] 12


Run
Result Size: 493 x 610
Get your own website
Women dataset
h <- factor(women$height)
print(h)
Random LETTERS sample
x <- factor(sample(LETTERS, 5))
print(x)
Manipulate factor
x <- factor(x, levels = LETTERS)
print(x)
Display levels
print(levels(x))
[1] 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72
Levels: 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72
[1] N Q D L E
Levels: D E L N Q
[1] N Q D L E
Levels: A B C D E F G H I J K L M N O P Q R S T U V W
[1] "A" "B" "C" "D" "E" "F" "G" "H" "I" "J" "K" "L"
[20] "T" "U" "V" "W" "X" "Y" "Z"


Run
Result Size: 493 x 610
Get your own website
Create a list
L <- list(
numbers = c(10, 20, 30),
matrix = matrix(1:4, nrow = 2),
fun = function(x) x * 2
)
Display content
print(L)
print(L$numbers)
print(L$matrix)
print(L$fun(5))
$numbers
[1] 10 20 30
$matrix
[,1] [,2]
[1,] 1 3
[2,] 2 4
$fun
function (x)
x * 2
[1] 10 20 30
[,1] [,2]
[1,] 1 3
[2,] 2 4
[1] 10

```r
. Factors of a number
- 12
 (i in 1:n) {
F (n %% i == 0) print(i)


. Generate 10 random integers between -50 and 50
- sample(-50:50, 10, replace = TRUE)
nt(x)

. FizzBuzz from 1 to 100
 (i in 1:100) {
f (i %% 15 == 0) print("FizzBuzz")
lse if (i %% 3 == 0) print("Fizz")
lse if (i %% 5 == 0) print("Buzz")
lse print(i)
```

```
[1] 1
[1] 2
[1] 3
[1] 4
[1] 6
[1] 12
 [1] -33  41  47   8  -3  46 -34  20  43  42
[1] 1
[1] 2
[1] "Fizz"
[1] 4
[1] "Buzz"
[1] "Fizz"
[1] 7
[1] 8
[1] "Fizz"
[1] "Buzz"
[1] 11
[1] "Fizz"
[1] 13
[1] 14
[1] "FizzBuzz"
[1] 16
[1] 17
[1] "Fizz"
[1] 19
[1] "Buzz"
[1] "Fizz"
[1] 22
[1] 23
[1] "Fizz"
[1] "Buzz"
```

Run

Result Size: 493 x 610

Get your own

```
lot(1, type="n", xlim=c(0, 20), ylim=c(0, 20),
    xlab="X-axis", ylab="Y-axis")
```



```r
exam_data <- data.frame(
  name = c("Ram", "Sam", "Raj"),
  score = c(85, 90, 75),
  attempts = c(1, 2, 3),
  qualifying = c(TRUE, TRUE, FALSE)
)

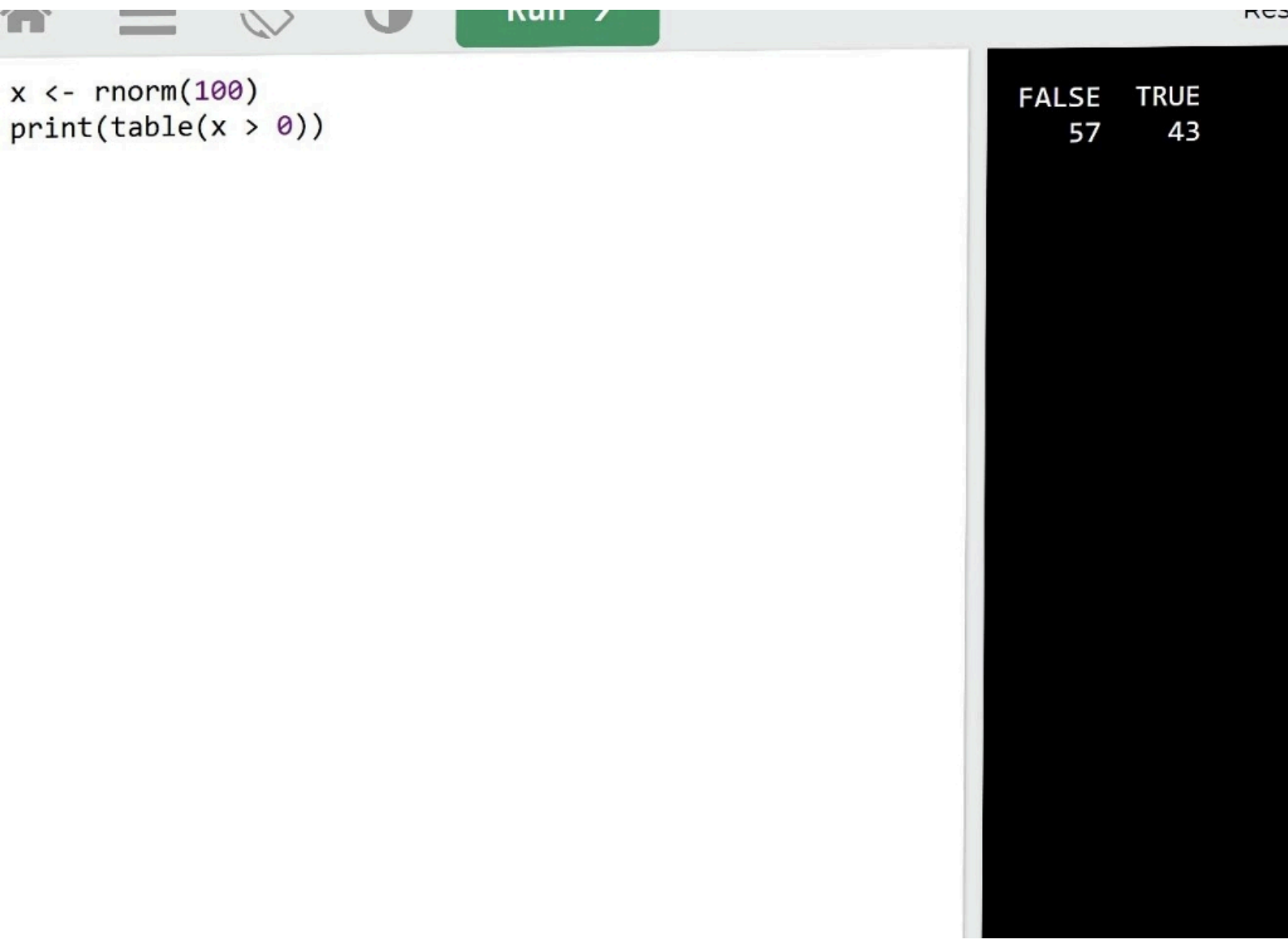
print(exam_data)
print(exam_data$name)

exam_data$grade <- c("A", "A+", "B")
exam_data <- rbind(exam_data, data.frame(
  name = "Tom", score = 80, attempts = 1,
  qualifying = TRUE, grade = "A"
))

exam_data <- exam_data[order(exam_data$score), ]
print(exam_data)

write.csv(exam_data, "exam_data.csv", row.names = FALSE)
```

```
  name score attempts qualifying
1  Ram    85        1        TRUE
2  Sam    90        2        TRUE
3  Raj    75        3       FALSE
[1] Ram Sam Raj
Levels: Raj Ram Sam
  name score attempts qualifying grade
3  Raj    75        3       FALSE     B
4  Tom    80        1        TRUE     A
1  Ram    85        1        TRUE     A
2  Sam    90        2        TRUE    A+
```


Run
x <- rnorm(100)
print(table(x > 0))
FALSE TRUE
57 43